

PRIMARY ISSUE CATEGORY: OVERPRESSURE / KICK

1. PROBLEM DEFINITION

A kick is an unplanned influx of formation fluid (gas, oil, or water) into the wellbore, occurring when formation pore pressure exceeds the hydrostatic pressure exerted by the drilling fluid column (i.e. the well is underbalanced).

2. EARLY WARNING SIGNS

- Pit volume gain not explained by surface operations.
- Flow observed from the well with pumps off (flow check positive).
- Increasing background and/or connection gas trend.
- Drilling break (sudden ROP increase) without a corresponding lithology change.

3. ROOT CAUSES (FIELD_ALPHA HISTORY)

- Pore pressure ramp underestimated in the pre-well prognosis for a given formation.
- Mud weight not increased ahead of a known or suspected overpressure transition.
- Swab pressure while tripping out too fast through a marginally balanced interval.
- Loss of hydrostatic head following an unrelated mud loss event.

4. STEP-BY-STEP SOLUTION / RESPONSE PROCEDURE

Step 1: Detect and confirm — Perform a flow check; if flowing, confirm a kick is in progress rather than a false alarm (e.g. thermal expansion).

Step 2: Shut in the well — Close the BOP (annular or ram) immediately using a hard or soft shut-in per the well control policy in effect.

Step 3: Record shut-in pressures — Record SIDPP and SICP and log pit gain; these feed the kill-sheet calculation.

Step 4: Calculate kill mud weight — Calculate kill mud weight from SIDPP and true vertical depth, and mix kill-weight mud on surface.

Step 5: Circulate out the kick — Circulate the influx out using the Driller's Method or Wait-and-Weight method per company policy, monitoring choke pressure throughout.

Step 6: Confirm well is dead — Confirm no-flow at zero pressure after circulation, then resume operations with the increased mud weight and continued gas monitoring.

5. MATERIALS / PARAMETERS QUICK REFERENCE

Parameter	Typical Value / Formula
Kill Mud Weight (SG)	Original MW + (SIDPP / (0.0981 x TVD in m))
Initial Circulating Pressure (ICP)	Slow pump pressure (SPP) + SIDPP
Final Circulating Pressure (FCP)	SPP x (Kill MW / Original MW)
Trip Margin (recommended)	0.03-0.05 SG above kill mud weight

6. PREVENTION CHECKLIST FOR OFFSET WELLS

- Run PWD/annular pressure-while-drilling tools ahead of any known or suspected pressure ramp.

- Begin a gradual, planned mud-weight increase before entering a suspected overpressure zone rather than relying on an emergency weight-up.
- Maintain a trip margin (typically 0.03-0.05 SG) above the estimated pore pressure gradient.
- Verify BOP function and float-valve integrity before drilling into any interval with kick risk.

7. FIELD CASE REFERENCES

Well	Depth	Outcome
W003	3210 m	Killed via Driller's Method, 2 circulations; MW raised 1.28 -> 1.42 SG

This Issue Resolution Playbook is a synthetically generated reference document produced for the NWIS prototype, consolidating lessons across Field_Alpha and Field_Alpha-adjacent wells. Not associated with an actual drilling operation.